

MENU

**Started on** Tuesday, 15 November 2022, 8:21 AM**State** Finished**Completed on** Tuesday, 15 November 2022, 9:59 AM**Time taken** 1 hour 37 mins

Question 1

Complete

Marked out of 6.00

The value (s) of k for which the matrix

$$\begin{bmatrix} -1 & 1 & k \\ 0 & 1 & 4 \\ k & 3 & 7 \end{bmatrix}$$

is singular is/are given by:

- ☐ $k = -1$ or $k = 2$
- ☐ $k = 5$ or $k = 1$
- ☐ $k = -1$ or $k = 0$
- ☐ None of the given option
- ☐ $k = 5$ and $k = 1$
- ☒ $k = -1$ or $k = 5$

Question 2

Complete

Not graded

Given two $n \times n$ invertible matrices A and B such that $(A - B)(B + A) = A^2 - B^2$. Which of the following statement is correct?

- ☒ None of the given options
- ☐ $B = A^{-1}BA$
- ☐ A is the inverses matrix of B
- ☐ $B = ABA^{-1}B$
- ☐ $B = BABA^{-1}$





Question 3

Marked out of 4.00

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Consider the matrices below with $A = AD$, where

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 3 & -1 & 1 & 3 \\ -2 & 5 & -k & k+1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 1 \\ 3 & -1 & 1 \\ -2 & 5 & k(k-1) \end{bmatrix}$$

For which value(s) of k will the entry X_{23} be zero?

- ☐ No such k
☐ $k = -2$
☒ $k = \frac{1 \pm \sqrt{5}}{2}$
☐ None of the given option
☐ $k = 0$ or $k = 1$
☐ $k = 3$

Question 4

Complete

Marked out of 4.00

Consider the system of linear equations with unknown a , b , and c and λ is parameter to be determined.

$$\begin{cases} a + (\lambda^2 + 1)b - (\lambda^2 - \lambda - 8)c = 3 - \lambda, \\ a - (\lambda^2 + 1)b = 3 \\ -(\lambda^2 + 1)b - (7 + \lambda - \lambda^2)c = \lambda. \end{cases}$$

Use elementary row operation to solve the above system. Find the value(s) for λ for which the system has no solution?

- ☒ $\lambda = -2$ and $\lambda = -3$
☐ $\lambda = 2$ or $\lambda = -3$
☐ $\lambda = -2$ or $\lambda = 3$
☐ None of the given options



Question 5

Complete

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Consider the matrices

$$X = \begin{bmatrix} x + y - 1 & 0 \\ 0 & 3 \end{bmatrix} \text{ and } Y = \begin{bmatrix} 5 & 0 \\ 0 & 2x - 3y - 7 \end{bmatrix}$$

The value(s) of y for which the matrices X^T and Y have no inverses is/are given by:

- ☐ $k = \frac{2}{5}$ and $y = k$
- ☐ No such value of y exists or $x = 2$
- ☐ $y = -2$
- ☒ $y = 1$

Question 6

Complete

Marked out of 5.00

Let C be an $m \times n$ matrix and k a scalar, prove that if $kC = O$ then either $k = 0$ or $C = O$ where O is the zero matrix with zero entries everywhere.

- ☒ Let $kC = O$, where $C = [C_{ij}]$ with $1 \leq i \leq m$ and $1 \leq j \leq n$. Then by identification, we have $kC_{ij} = 0$ for $1 \leq i \leq m$ and $1 \leq j \leq n$
- ☒ This implies that $k = 0$ or $C = O$ for $1 \leq i \leq m$ and $1 \leq j \leq n$
- ☐ If there exist i or j such that $C_{ij} \neq 0$ Then $k = 0$ which implies that $kC = O$.
- ☐ Otherwise, $C_{ij} = 0$ for $1 \leq i \leq m$ and $1 \leq j \leq n$ which implies that $C = O$ is indeed the zero matrix. This conclude the desired proof.
- ☐ None of the given options is correct



Question 7

Complete

Not graded

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Let A be the coefficient matrix of the system

$$3x + 4y + z = 8$$

$$-2x + 7y + z = 1$$

$$8x + y + 5z = -2.$$

Now add the following cofactor entries (of A) C_{32} , C_{23} and C_{22} to get the resulting sum:

- ☐ 31
- ☐ -3
- ☐ 0
- ☐ -1
- ☒ None of the given options

Question 8

Complete

Marked out of 3.00

Let A be square matrix of the same size. Which of the following statement is incorrect?

- ☐ $(A^T + A)^T$ is skew-symmetric
- ☐ $A + A^T$ is symmetric
- ☒ None of the given options
- ☐ AA^T is symmetric
- ☐ $(A^T - A)^T$ is symmetric and $A + A^T$ is skew-symmetric



Question 9

Complete

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Verify whether or not the matrix below in reduced row echelon.

$$\begin{bmatrix} 1 & 10 & 0 & 3 & 0 \\ 0 & 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- ☐ TRUE and FALSE
- ☒ TRUE or FALSE
- ☐ None of the given option
- ☐ FALSE
- ☒ TRUE

Question 10

Complete

Marked out of 3.00

Use inspection to find the determinant of $C = AB$, where

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

- ☒ -1
- ☐ 2
- ☐ 30
- ☐ None of the given options
- ☐ 3



Question 11

Complete

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Assume that A is a 2×2 matrix with integer entries. If B is the inverse of A and B has integer entries, then choose the correct statement.

- ☐ $A^2 = I$, where I is the corresponding identity matrix.
- ☐ None of the given option
- ☐ $\det(B) = 0$.
- ☐ $A = 2I$ or $A = B$, where I is the corresponding identity matrix.
- ☒ $\det(B) = -2$

Question 12

Complete

Marked out of 6.00

Suppose L is the line with symmetric equations

$$x - 3 = y - 1 = -z + 2$$

Now determine whether the above line is parallel to the plane

$$4x - y + 2z = 1.$$

Choose the correct answer.

- ☒ L is not parallel to the given plan
- ☐ L is perpendicular to the vector $(1, 1, -1)^T$ where T stands for the transpose
- ☐ None of the given option
- ☐ L is not perpendicular to the given plan
- ☐ L is parallel to the given plan



Question 13

Complete

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The augmented matrices of the three systems are given below.

$$(i) \left[\begin{array}{ccc|c} 1 & a & 0 & 4 \\ 1 & 0 & 1 & -2 \\ 0 & 0 & b & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \text{ where } a, b \in \mathbb{R} \text{ with } b \neq 0.$$

How many solutions are there in each case?

- ☒ (i) infinitely many solutions (ii) Exactly one solution (iii) No solutions
- ☐ (i) Exactly one solution (ii) infinitely many solutions (iii) No solutions
- ☐ (i) Exactly one solution (ii) No solutions (iii) infinitely many solutions
- ☐ (i) No solution (ii) infinitely many solutions (iii) Unique solutions
- ☐ None of the given options

Question 14

Complete

Marked out of 3.00

We consider the two points $P(1, 2, 1)$ and $Q(2, -1, 1)$.

Determine the symmetric equations of the line passing through the points P and Q . Assume that $R(3, -4, -3)$ is another point. Assume you want to show using the symmetric equations of the line that the three points are collinear.

Find the value (s) for the parameter t for which $R = P + tV$ implies the three points are collinear, where V is tip of the vector PQ .

- ☐ None of the given options
- ☐ $t = 0$ or $t = -1$
- ☒ $t = 2$ and $t = -1$
- ☐ $t = -2$ or $t = 1$
- ☐ $t = 2$



Question 15

Complete

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Find the symmetric equations of the line L that passes through the point $P(-1, 2, -3)$ and is parallel to the vector $\vec{U} = \langle 1, -2, -1 \rangle$. Find the point which the required line L intersects with the coordinate planes.

- ☐ The point of intersection of L with the yz -plane is $(0, 0, -4)$.
- ☒ The point of intersection of L with the yz -plane is $(0, 1, 4)$.
- ☐ None of the given options
- ☐ The point of intersection of L with the yz -plane is $(0, 0, -3)$.
- ☐ The point of intersection of L with the yz -plane is $(0, 0, 4)$.

Question 16

Complete

Marked out of 6.00

Let V be the plane passing through the points $R = (2, -1, 1)$, $S = (1, 2, -1)$ and $T = (-3, 1, -2)$. Let U be the plane perpendicular to V and passing through the points S and $(0, 1, 1)$.

The cosine of the angle between U and V is given by:

- ☒ $\frac{1}{\sqrt{2}}$
- ☐ 0
- ☐ 1
- ☐ None of the given options
- ☐ -1



Question **17**

Complete

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Given three points $A(1, 0, -1)$, $B(0, 2, 1)$, $C(1, 1, -1)$. The area of the triangle ABC is given by:

- ☐ None of the given options
- ☐ 8 units^2
- ☐ 4 units^2
- ☐ $\frac{3}{2} \text{ units}^2$
- ☒ 3 units^2

Question **18**

Complete

Marked out of 5.00

Simplify the complex number

$$i^{1387}$$

Choose the correct answer:

- ☐ i
- ☒ 1
- ☐ $-i$
- ☐ None of the given options
- ☐ -1



Question **19**

Complete

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Let

$$z_1 = 1 - 2i - i^2$$

and

$$z_2 = x + y + ix$$

be two complex numbers such that

$$z_1 = z_2.$$

Solve for x and y from which you deduce that the product xy is equal to

- ☒ None of the given options
- ☐ -2
- ☐ 8
- ☐ 4
- ☐ -8

Question **20**

Complete

Not graded

Let L be the line that passes through the points $A(-1, 1, -1)$ and $B(1, -1, 2)$, and V the plane containing L and the point $C(1, 0, 1)$. The parallel vectors $\vec{u}$ and $\vec{v}$ lying in the plane V are given by:

- ☐ None of the given options
- ☐ $\vec{u} = (2, -2, 3)$ and $\vec{v} = (2, -1, -1)$
- ☐ $\vec{u} = (2, -2, 3)$ and $\vec{v} = (2, 1, 2)$
- ☐ $\vec{u} = (1, -2, 3)$ and $\vec{v} = (2, -1, 2)$
- ☒ $\vec{u} = (2, -2, 3)$ and $\vec{v} = (2, -1, 2)$



Question 21

Complete

Not graded

MENU

Let L be the line that passes through the points $A(-1, 1, -1)$ and $B(1, -1, 2)$, and V the plane containing L and the point $C(1, 0, 1)$. First, determine two parallel vectors $\vec{u}$ and $\vec{v}$ that lying in the plane V . Then, the Cartesian equation of the plane is given by:

- ☐ None of the given options
- ☐ $x - 2y - 2z - 1 = 0$
- ☒ $x - 2y + 2z - 1 = 0$
- ☐ $x + 2y - 2z - 1 = 0$
- ☐ $x - 2y - 2z + 1 = 0$

Question 22

Complete

Not graded

Let $P(z) = z^2 - 2z + 4$ and assume that the root of $P(z)$ are given by α and β . The expression for $\alpha^n + \beta^n$ is given by

- ☐ 4 for $n = 0$
- ☐ 2 for $n = 0$
- ☒ None of the given options
- ☐ $-2i$ for $n = 0$
- ☐ -2 for $n = 0$



Question **23**

Complete

Marked out of 2.00

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Solve $z^3 - 8 = 0$. The roots are given by.

- ☐ None of the given option
- ☐ $(z_0 = 2 \left[\cos \pi + i \sin \pi \right], z_1 = 2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right], z_2 = 2 \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right])$
- ☐ $(z_0 = 2 \left[\cos 0 + i \sin 0 \right], z_1 = 2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right], z_2 = 2 \left[\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right])$
- ☐ $(z_0 = 2 \left[\cos \pi - i \sin \pi \right], z_1 = 2 \left[\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right], z_2 = 2 \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right])$
- ☒ $(z_0 = 2 \left[\cos 0 + i \sin 0 \right], z_1 = 2 \left[\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right], z_2 = 2 \left[\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right])$

Question **24**

Complete

Marked out of 3.00

Let

$$z = 4 + 3i$$

Then $|z|^2$ is given by

- ☐ 5
- ☐ 6
- ☐ None of the given options
- ☐ 3
- ☒ 25



Question **25**

Complete

Not graded

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Consider that z is a real number, determine the value(s) for β such that

$$z = \frac{\beta + 2i}{1 + i\beta}$$

- ☒ $\beta = \pm 1$
- ☐ $\beta = \pm 3$
- ☐ $\beta = \pm 2$
- ☐ None of the given options
- ☐ $\beta = \pm \sqrt{2}$

Question **26**

Complete

Not graded

Find the general form of the matrix that commute with the matrix $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$

- ☒ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a = 2d$
- ☐ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a \neq d$
- ☐ $\begin{bmatrix} a & a-d \\ 0 & 2d \end{bmatrix}$ whenever $a = d$



Question **27**

Complete

Not graded

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Assume that $C = A - I_3$ where

$$A = \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & -1 \end{bmatrix}.$$

Find the matrix B such that

$$CB^T = 3 \begin{bmatrix} -1 & 2 & 0 \\ 3 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

where I_3 is the 3 by 3 identity matrix.

- ☐ None of the given options
- ☒ $B = \begin{bmatrix} 3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐ $B = \begin{bmatrix} -3 & 4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$
- ☐ $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & 6 \end{bmatrix}$
- ☐ $B = \begin{bmatrix} -3 & -4 & 3 \\ 9 & 2 & -6 \end{bmatrix}$

◀ Additional Resources

Jump to...

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